



Retriever Performance Report

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Dit rapport vergelijkt 4 retrievers voor retrieval-augmented generation. Het geeft een samenvatting van de corpus- en queryset, vergelijkt de rankingkwaliteit en runtime, en toont vervolgens de configuratie en voorbeeldresultaten van elke retriever. Gebruik de vergelijkingsectie om te bepalen welke retriever het meest waarschijnlijk nuttig bewijs in het contextvenster van de generator plaatst.

De gesynthetiseerde retrieval-dataset is ontworpen om het prestatieniveau van retrieval-methoden binnen Retrieval-Augmented Generation (RAG) te evalueren. Het corpus bevat technische tekstdocumenten of tekstchunks die gerelateerd zijn aan frameworks zoals Symfony, met een focus op onderwerpen zoals messaging, dependency injection, routing en event handling. Deze documenten bevatten gedetailleerde uitleg over configuraties, implementaties en probleemoplossingen. De synthetisch gegenereerde vragen zijn realistische zoekopdrachten van developers die zich afvragen waarom bepaalde functionaliteiten niet werken, zoals fouten bij autowiring, problemen met event listeners of issues bij het normaliseren van data. De vragen zijn ontworpen om troubleshooting- en configuratiegerelateerde problemen te benadrukken, en vormen een echte reflectie van de problemen die developers in de praktijk tegenkomen. De dataset kan worden gebruikt om de efficiëntie en relevantie van retrieval-systemen te meten, door het beoordelen van hoe goed een systeem de juiste documenten kan vinden op basis van de vragen. Hierbij wordt gebruik gemaakt van een strakke relevante beoordelingsmethode die op basis van de daadwerkelijke problemen en technische contexten bepaalt of een document nuttig is voor het oplossen van de vraag.

Machine Summary

Machine Summary

System	Linux
Processor	x86_64
Cores	32
Ram (Gb)	377.56
Python Version	3.12.3
Cuda	Available (2 GPU(s))
Gpu Name	NVIDIA RTX A5000

Corpus Summary

Corpus and Queries

Documents	2,594
Documents With Title	0
Avg Document Chars	695.1766
Min Document Chars	5
Max Document Chars	1,000
Queries	52
Avg Query Chars	67.9231
Min Query Chars	45

Max Query Chars	125
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Sample Query

- why are my assets not being versioned correctly in production
- why is my application test not reaching the controller endpoint
- why are my custom validation constraints not being triggered
- why is my request object not containing the form data
- why is my serializer not applying the correct date format for normalization

Retriever Comparison

Retrieval and Rerank Metrics

Retriever	NDCG@10	Recall@10
BM25Retriever (bm25)	0.5664	0.601
BM25Retriever (bm25) + rerank	0.7265	0.6798
DenseRetrieverSentenceBert (dense)	0.7858	0.8279
DenseRetrieverSentenceBert (dense) + rerank	0.8388	0.832
HybridRetriever (hybrid)	0.7869	0.8288
HybridRetriever (hybrid) + rerank	0.8388	0.832
LinearRagRetriever (linear)	0.7892	0.7894
LinearRagRetriever (linear) + rerank	0.8165	0.8022

Chart Colors

Series
Base retrieval
Rerank



Uitleg metrics: Deze scores evalueren de retrievalfase voordat het taalmodel een antwoord genereert. Binnen een RAG-pipeline betekent betere retrieval dat de generator relevanter en beter gerankt bewijs ontvangt en minder hoeft te vertrouwen op niet-onderbouwde modelkennis.

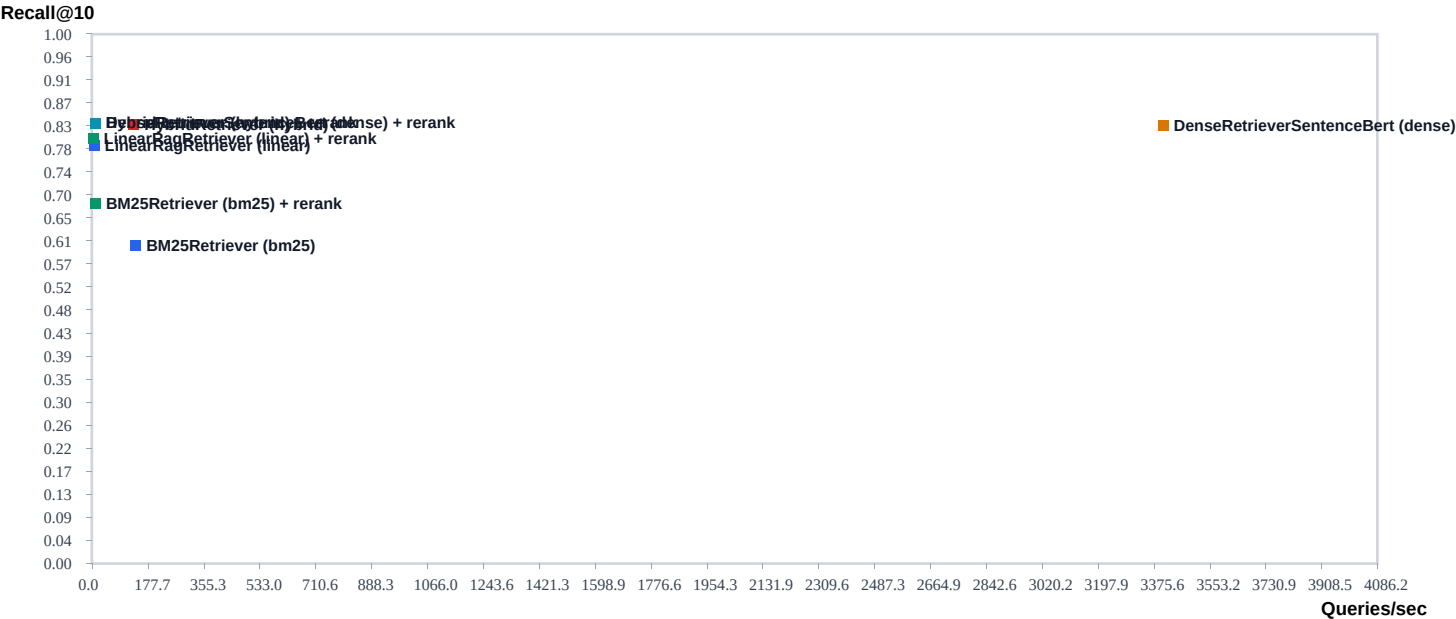
Recall@k meet dekking: hoeveel relevante documenten zijn gevonden binnen de top k. **Precision@k** meet focus: hoeveel van de top k daadwerkelijk relevant is in plaats van afleidende context. **MAP@k** belooft relevante documenten die consistent vroeg verschijnen over meerdere queries heen. **NDCG@k** belooft zowel de rangorde als graduele relevantie en is daarom vaak de beste enkele indicator voor de vraag of het sterkste bewijs als eerste de prompt bereikt.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de hoogste score te kiezen, maar om te bepalen welke retriever het beste past bij het doel van de RAG-toepassing. Een hoge Recall@k is vooral belangrijk wanneer het systeem zo min mogelijk relevante informatie mag missen, bijvoorbeeld bij juridische of ondersteunende zoekvragen. Een hoge Precision@k is belangrijk wanneer het contextvenster beperkt is en irrelevante passages de generator kunnen afleiden. Wanneer reranking duidelijk betere NDCG- of MAP-scores geeft, betekent dit dat de juiste documenten niet alleen gevonden worden, maar ook hoger in de ranglijst komen te staan. Dat is praktisch waardevol omdat een taalmodel meestal sterker leunt op de eerste passages in de context. Runtime moet daarnaast worden meegewogen: een iets lagere score kan acceptabel zijn wanneer de retriever veel sneller is of eenvoudiger te onderhouden blijft.

Practical Limitations and Runtime Trade-offs

Retriever	Query Time	Queries/Sec	Index Time	Document/Sec	Storage
BM25Retriever (bm25)	0.3729s	139.4558	6.4667s	401.1343	2.05MB
BM25Retriever (bm25) + rerank	4.6208s	11.2523	6.4667s	401.1343	2.05MB
DenseRetrieverSentenceBert (dense)	0.0153s	3405.164	38.9325s	66.6282	10.43MB
DenseRetrieverSentenceBert (dense) + rerank	4.3665s	11.9057	38.9325s	66.6282	10.43MB
HybridRetriever (hybrid)	0.3893s	133.566	45.3991s	57.1377	12.49MB
HybridRetriever (hybrid) + rerank	4.7374s	10.9785	45.3991s	57.1377	12.49MB
LinearRagRetriever (linear)	6.8462s	7.5954	48.0415s	53.995	2.29GB
LinearRagRetriever (linear) + rerank	11.0878s	4.6891	48.0415s	53.995	2.29GB

Quality versus Query Throughput

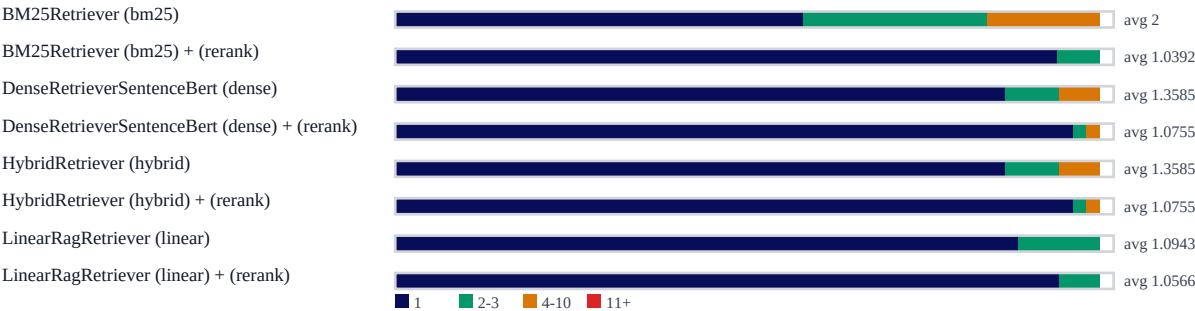


Uitleg runtime en throughput: Deze resultaten evalueren de praktische prestaties van de retrievalfase naast retrievalkwaliteit. Query time meet hoeveel tijd nodig is om queries te verwerken, terwijl index time aangeeft hoeveel tijd nodig is om de volledige index op te bouwen. Queries/sec en documents/sec meten de throughput van het systeem en geven inzicht in schaalbaarheid binnen grotere RAG-omgevingen. Storage MB toont hoeveel opslagruimte de retrievalindex gebruikt, wat belangrijk kan zijn binnen productieomgevingen waar infrastructuurkosten en geheugenverbruik een rol spelen. De grafiek combineert retrievalkwaliteit met throughput door Recall@10 af te zetten tegen queries per seconde. Hogere posities betekenen betere retrievaldekking, terwijl posities verder naar rechts wijzen op hogere snelheid en lagere latency.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de snelste of meest accurate retriever te kiezen, maar om te bepalen welke retrievalarchitectuur het beste aansluit bij de praktische eisen van de toepassing. Een retriever rechtsboven in de grafiek combineert hoge retrievalkwaliteit met hoge throughput en vormt daardoor meestal de beste algemene balans voor real-time RAG-systemen. Een retriever kan echter bewust meer querytijd gebruiken om betere ranking of hogere retrievaldekking te behalen. In productieomgevingen moet daarom rekening worden gehouden met latency, schaalbaarheid, hardwarekosten en onderhoudscomplexiteit. Een iets lagere retrievalscore kan acceptabel zijn wanneer een systeem aanzienlijk sneller, goedkoper of eenvoudiger schaalbaar blijft.

Rank Distribution

Retriever	Queries	Rank 1	Rank 2-3	Rank 4-10	Rank 11+	Avg First Rank
BM25Retriever (bm25)	51	29	13	8	0	2
BM25Retriever (bm25) + (rerank)	51	47	3	0	0	1.0392
DenseRetrieverSentenceBert (dense)	53	45	4	3	0	1.3585
DenseRetrieverSentenceBert (dense) + (rerank)	53	50	1	1	0	1.0755
HybridRetriever (hybrid)	53	45	4	3	0	1.3585
HybridRetriever (hybrid) + (rerank)	53	50	1	1	0	1.0755
LinearRagRetriever (linear)	53	46	6	0	0	1.0943
LinearRagRetriever (linear) + (rerank)	53	49	3	0	0	1.0566

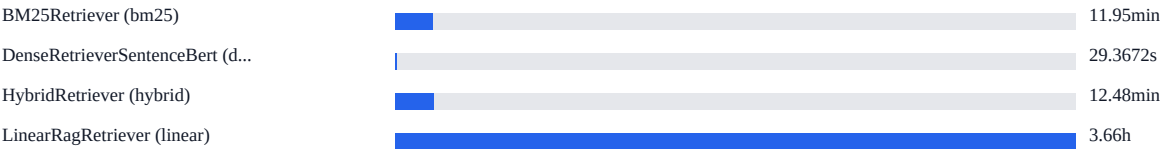


Uitleg rank distribution: Rank distribution laat zien op welke positie het eerste relevante document verschijnt voor iedere query. In plaats van alleen een algemene retrievalscore te tonen, maakt deze verdeling zichtbaar hoe snel bruikbare informatie beschikbaar komt voor de RAG-pipeline. Rank 1 betekent dat direct het eerste resultaat relevant is. Rank 2-3 betekent dat relevante informatie nog steeds zeer vroeg wordt gevonden. Rank 4-10 geeft aan dat de retriever dieper in de ranking moet zoeken voordat bruikbaar bewijs verschijnt. Rank 11+ betekent dat relevante informatie pas laat wordt gevonden, waardoor de kans groter wordt dat het taalmodel minder sterke of minder gefocuste context ontvangt. De gemiddelde first rank vat samen hoe vroeg het eerste relevante document gemiddeld verschijnt; lagere waarden duiden op sterkere rankingprestaties.

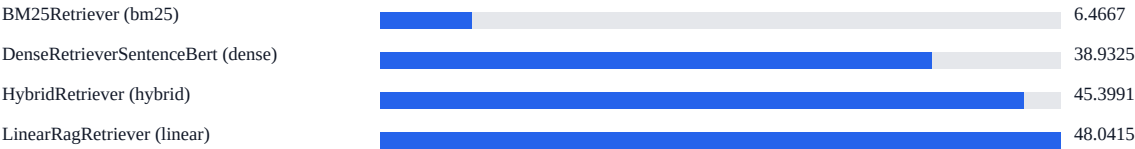
Praktische interpretatie: Een retriever met veel Rank 1 en Rank 2-3 resultaten levert doorgaans directer bruikbaar bewijs voor generatie. Dit is vooral waardevol wanneer het contextvenster beperkt is of wanneer de generator sterk afhankelijk is van de eerste passages in de prompt. Een groter aandeel Rank 4-10 kan nog steeds bruikbaar zijn, maar betekent dat relevante informatie minder prominent wordt aangeboden. Een groot Rank 11+ aandeel wijst vaak op zwakkere rankingkwaliteit of op een grotere behoefte aan reranking, hybride retrieval of betere queryverwerking. Rank distribution helpt daarom niet alleen te bepalen of relevante documenten worden gevonden, maar vooral hoe snel en hoe zichtbaar deze documenten voor de generator beschikbaar zijn.

Scalability Estimates

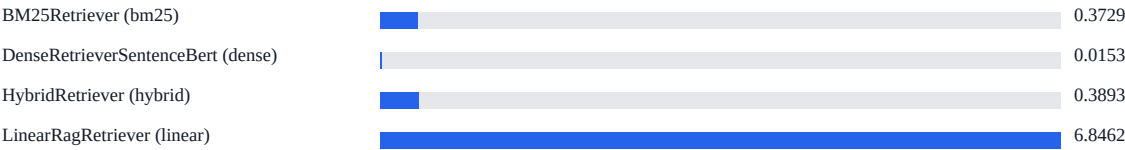
Retriever	Target Docs	Est. Index Time	Target Queries	Est. Query Time	Est. Queries/Sec	Est. Storage
BM25Retriever (bm25)	1,000,000	41.55min	100,000	11.95min	139.4558	791.60MB
DenseRetrieverSentenceBert (dense)	1,000,000	4.17h	100,000	29.3672s	3405.164	3.93GB
HybridRetriever (hybrid)	1,000,000	4.86h	100,000	12.48min	133.566	4.70GB
LinearRagRetriever (linear)	1,000,000	5.14h	100,000	3.66h	7.5954	881.85GB



Total Index Time (seconds) (2594 docs)



Total Query Time (seconds) (52 queries)

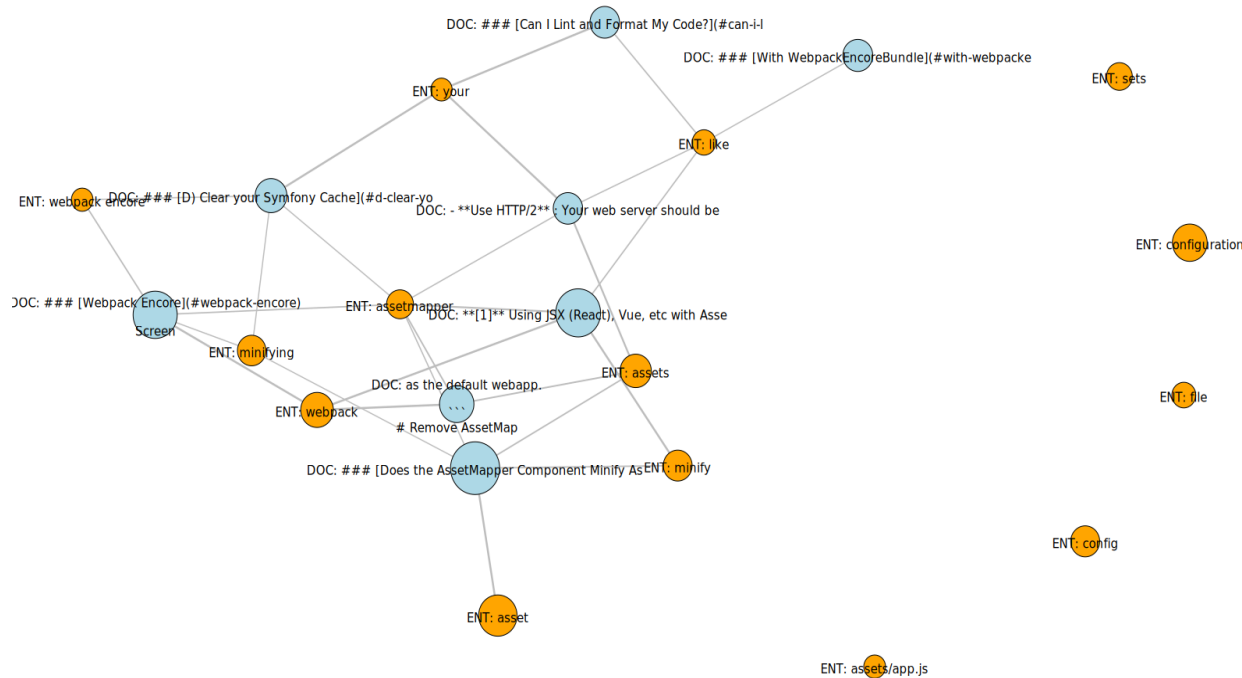


Rerank Time (seconds) (20 - docs / 52 - queries)



Query Graph - LinearRagRetriever (linear)

Query: why is my webpack configuration not minifying the assets



Retrieved Documents

Rank	Document	Score	Text
1	ece57511-4f43-435e-af7e-34db3c4a4b53	0.0164	### [Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is...
2	38e1c5f2-ef8d-4104-b73e-a8b100d7029d	0.0161	**[1]** Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation...
3	657d9e76-c5fb-454a-b6c3-e8574a48b26c	0.0159	### [Webpack Encore](#webpack-encore) Screencast Do you prefer video tutorials? Check out the [Webpack Encore screencast series...
4	a50a3430-d27e-4632-9405-b3b1da42528c	0.0156	as the default webapp. `` # Remove AssetMapper & Turbo/Stimulus temporarily \$ composer remove symfony/ux-turbo symfony/asset-m...
5	9bcb2b7a-a36c-4354-a5a1-a67c54c25450	0.0154	### [D] Clear your Symfony Cache](#d-clear-your-symfony-cache) Make sure you clear and warm-up your Symfony cache: `` \$ APP_EN...

6	5a2c522a-a5a3-40ae-9aab-72c6b8a8dc10	0.0152	### [With WebpackEncoreBundle](#with-webpackencorebundle) If you're using Webpack Encore, the recipe will also update your `web...
7	b57212bb-dfee-4650-9f72-57fa2cb8c67d	0.0149	### [Can I Lint and Format My Code?](#can-i-lint-and-format-my-code) Not with AssetMapper, but you can install [kocal/biome-js-...
8	715e07f2-3eea-4bce-bcde-c1bafa77f2dd	0.0147	- **Use HTTP/2** : Your web server should be running HTTP/2 or HTTP/3 so the browser can download assets in parallel. HTTP/2 is...

BM25Retriever (bm25)

Configuration

Retriever Type	bm25
Topk	20
Language	english
Index Name	raccoon-bm25-test
Content Field	content
Metadata Field	metadata
Refresh On Write	yes
Timeout	120

Result Summary

Queries	52
Total Results	1,040
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	6.0358
Avg Score	12.7854
Max Score	36.5253

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.5321	0.4983	0.4895	0.5664	0.6117
MAP	0.1314	0.2736	0.3589	0.4266	0.4554
Recall	0.1314	0.3086	0.45	0.601	0.7202
Precision	0.5577	0.4679	0.4192	0.2846	0.1721

Retriever Metrics

Index Time / Indexing / Time In Seconds	6.4667
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Docs Per Second	401.1343
Index Time / Indexing / Backend	elasticsearch
Index Time / Indexing / Storage Size Bytes	2,153,151
Index Time / Indexing / Storage Size Mb	2.0534
Query Time / Search / Time In Seconds	0.3729
Query Time / Search / Time Per Query In Seconds	0.0072
Query Time / Search / Queries Per Second	139.4558
Query Time / Search / Queries	52

Query Time / Search / Top K	20
Query Time / Search / Backend	elasticsearch
Query Time / Search / Total Results	1,040
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	6.0358
Query Time / Search / Max Score	36.5253
Query Time / Search / Avg Score	12.7854

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	52
Total Pairs	1,040
Total Batches	104
Total Input Tokens	251,926
Total Wall Time Sec	4.2479
Avg Time Query Sec	0.0817
Avg Time Batch Sec	0.0408
Pairs Per Sec	244.83
Queries Per Sec	12.24
Tokens Per Sec	59306.36
Peak Gpu Memory Gb	1.21

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.875	0.7676	0.6913	0.7265	0.7422
MAP	0.2016	0.4564	0.5787	0.6192	0.6298
Recall	0.2016	0.4686	0.6067	0.6798	0.7202
Precision	0.9039	0.7372	0.5769	0.325	0.1721

Sample Top Results

Query q0: why is my webpack configuration not minifying the assets

1. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 20.6974

****[1]**** Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. ****[2]**** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets

with AssetMapper. **[3]** If you use `npm`, there are update chec...

2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 18.1002

[Webpack Encore](#webpack-encore) Screencast Do you prefer video tutorials? Check out the [Webpack Encore screencast series](https://symfonycasts.com/screencast/webpack-encore) . [Webpack Encore](https://www.npmjs.com/package/@symfony/webpack-encore) is a simpler way to integrate [Webpack](https://webpack.js.org/) into your application. It wraps Webpack, giving you a clean & powerful API for bundling JavaScript modules, pre-processing CSS & JS and compiling and minifying assets. [...]

3. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score 17.7176

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

Sample Top Rerank Results

Query q0: why is my webpack configuration not minifying the assets

1. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0904

[1] Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. **[2]** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets with AssetMapper. **[3]** If you use `npm`, there are update chec...

2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score -0.1967

[Webpack Encore](#webpack-encore) Screencast Do you prefer video tutorials? Check out the [Webpack Encore screencast series](https://symfonycasts.com/screencast/webpack-encore) . [Webpack Encore](https://www.npmjs.com/package/@symfony/webpack-encore) is a simpler way to integrate [Webpack](https://webpack.js.org/) into your application. It wraps Webpack, giving you a clean & powerful API for bundling JavaScript modules, pre-processing CSS & JS and compiling and minifying assets. [...]

3. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score -0.3291

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

DenseRetrieverSentenceBert (dense)

Configuration

Retriever Type	dense
Topk	20
Model Id	snowflake/snowflake-arctic-embed-l-v2.0
Max Length	512
Device	cuda
Query Prompt Name	query
Passage Prompt Name	document
Normalize Embeddings	no
Batch Size	128
Corpus Chunk Size	50,000
Query Chunk Size	1,024
Show Progress Bar	yes

Result Summary

Queries	52
Total Results	1,040
Avg Results/Query	20
Min Results/Query	20

Max Results/Query	20
Min Score	0.2388
Avg Score	0.4563
Max Score	0.7895

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.8077	0.7228	0.6837	0.7858	0.8297
MAP	0.1939	0.424	0.5608	0.6693	0.7005
Recall	0.1939	0.4414	0.6205	0.8279	0.9365
Precision	0.8654	0.6859	0.5846	0.3942	0.224

Retriever Metrics

Index Time / Encoding / Time In Seconds	38.9325
Index Time / Encoding / Documents	2,594
Index Time / Encoding / Queries	52
Index Time / Encoding / Shards	1
Index Time / Encoding / Embedding Dim	1,024
Index Time / Encoding / Size In Bytes	10,939,836
Index Time / Encoding / Size In Mb	10.433
Index Time / Encoding / Docs Per Second	66.6282
Query Time / Search / Time In Seconds	0.0153
Query Time / Search / Time Per Query In Seconds	0.0003
Query Time / Search / Queries Per Second	3405.164
Query Time / Search / Queries	52
Query Time / Search / Top K	20
Query Time / Search / Score Function	dot
Query Time / Search / Use Faiss	no
Query Time / Search / Backend	torch
Query Time / Search / Total Results	1,040
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.2388
Query Time / Search / Max Score	0.7895
Query Time / Search / Avg Score	0.4563

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	52
Total Pairs	1,040
Total Batches	104
Total Input Tokens	246,206
Total Wall Time Sec	4.3512
Avg Time Query Sec	0.0837
Avg Time Batch Sec	0.0418
Pairs Per Sec	239.01
Queries Per Sec	11.95
Tokens Per Sec	56583.52
Peak Gpu Memory Gb	3.32

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9263	0.8348	0.7758	0.8388	0.8777
MAP	0.2131	0.5062	0.6712	0.7417	0.7715
Recall	0.2131	0.518	0.6987	0.832	0.9365
Precision	0.9615	0.8077	0.6654	0.3981	0.224

Sample Top Results

Query q0: why is my webpack configuration not minifying the assets

1. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score 0.6359

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

2. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.5996

[1] Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. **[2]** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets with AssetMapper. **[3]** If you use `npm`, there are update chec...

3. Document a50a3430-d27e-4632-9405-b3b1da42528c | score 0.492

as the default webapp. ``` # Remove AssetMapper & Turbo/Stimulus temporarily \$ composer remove symfony/ux-turbo symfony/asset-mapper symfony/stimulus-bundle # Add Webpack Encore & Turbo/Stimulus back \$ composer require symfony/webpack-encore-bundle symfony/ux-turbo symfony/stimulus-bundle # Install & Build Assets \$ npm install \$ npm run dev ```

Sample Top Rerank Results

Query q0: why is my webpack configuration not minifying the assets

1. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0904

****[1]**** Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. ****[2]**** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets with AssetMapper. ****[3]**** If you use `npm`, there are update chec...

2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score -0.1967

[Webpack Encore](#webpack-encore) Screencast Do you prefer video tutorials? Check out the [Webpack Encore screencast series](https://symfonycasts.com/screencast/webpack-encore) . [Webpack Encore](https://www.npmjs.com/package/@symfony/webpack-encore) is a simpler way to integrate [Webpack](https://webpack.js.org/) into your application. It wraps Webpack, giving you a clean & powerful API for bundling JavaScript modules, pre-processing CSS & JS and compiling and minifying assets. [...]

3. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score -0.3291

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

HybridRetriever (hybrid)

Configuration

Retriever Type	hybrid
K	60
Retrievers	bm25:0.3, dense:0.7

Result Summary

Queries	52
Total Results	1,040
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0087
Avg Score	0.0117
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.8333	0.7278	0.6779	0.7869	0.8319
MAP	0.1939	0.4311	0.5595	0.6658	0.7024
Recall	0.1939	0.449	0.6135	0.8288	0.9365
Precision	0.8654	0.6987	0.5808	0.3942	0.224

Retriever Metrics

Query Time / Search / Time In Seconds	0.3893
Query Time / Search / Retriever Time In Seconds	0.3881
Query Time / Search / Rrf Time In Seconds	0.0012
Query Time / Search / Wrapper Time In Seconds	0.0013
Query Time / Search / Execution Mode	sequential

Query Time / Search / Time Per Query In Seconds	0.0075
Query Time / Search / Queries Per Second	133.566
Query Time / Search / Queries	52
Query Time / Search / Top K	20
Query Time / Search / Retrievers	2
Query Time / Search / Rrf K	60
Query Time / Search / Total Results	1,040
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.0087
Query Time / Search / Max Score	0.0164
Query Time / Search / Avg Score	0.0117
Index Time / Indexing / Time In Seconds	45.3991
Index Time / Indexing / Documents	5,188
Index Time / Indexing / Docs Per Second	114.2753
Index Time / Indexing / Retrievers	2
Index Time / Indexing / Size In Mb	12.4864

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	52
Total Pairs	1,040
Total Batches	104
Total Input Tokens	246,206
Total Wall Time Sec	4.3481
Avg Time Query Sec	0.0836
Avg Time Batch Sec	0.0418
Pairs Per Sec	239.19
Queries Per Sec	11.96
Tokens Per Sec	56624.32
Peak Gpu Memory Gb	3.32

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9263	0.8348	0.7758	0.8388	0.8777
MAP	0.2131	0.5062	0.6712	0.7417	0.7715
Recall	0.2131	0.518	0.6987	0.832	0.9365
Precision	0.9615	0.8077	0.6654	0.3981	0.224

Sample Top Results

Query q0: why is my webpack configuration not minifying the assets

1. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score 0.0162

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

2. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0162

[1] Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. **[2]** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets with AssetMapper. **[3]** If you use `npm` , there are update chec...

3. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 0.0158

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Sample Top Rerank Results

Query q0: why is my webpack configuration not minifying the assets

1. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0904

[1] Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation. Also, some features (like Vue single-file components) cannot be compiled down to pure JavaScript that can be executed by a browser. **[2]** You can install the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) to minify CSS/JS code (and remove all comments) when compiling assets with AssetMapper. **[3]** If you use `npm` , there are update chec...

2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score -0.1967

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3. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score -0.3291

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LinearRagRetriever (linear)

Configuration

Retriever Type	linear
Embedding Model Name	snowflake/snowflake-arctic-embed-l-v2.0
Spacy Model Name	nl_core_news_sm
Dataset Name	symphony_beir_semantic
Cache Path	data/processed/symfony/symphony_beir_semantic/linear_rag_cache
Device	cuda
Max Seq Length	512

Embed Batch Size	128
Ner Batch Size	64
Retrieval Top K	20
Dense Candidate K	500
Bm25 Candidate K	500
Graph Candidate K	500
Local Graph Dense Seed K	200
Local Graph Bm25 Seed K	200
Local Graph Sentence Seed K	50
Local Graph Concept Seed K	50
Dense Rrf Weight	0
Bm25 Rrf Weight	0
Graph Rrf Weight	1
Rrf K	60
Min Concept Len	4
Max Concept Words	4
Min Concept Df	2
Max Concept Df Ratio	0.1
Ppr Damping	0.6
Ppr Max Result Docs	500
Export Query Graph	yes
Use Gpu For Spacy	yes

Result Summary

Queries	52
Total Results	1,040
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0125
Avg Score	0.0143
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@50	@100
NDCG	0.8333	0.7774	0.7211	0.7892	0.8345	0.8345
MAP	0.2003	0.4636	0.5926	0.6689	0.7064	0.7064
Recall	0.2003	0.4888	0.6484	0.7894	0.9058	0.9058
Precision	0.8846	0.7564	0.6115	0.375	0.0865	0.0433

Retriever Metrics

Index Time / Indexing / Time In Seconds	48.0415
Index Time / Indexing / Documents	2,594

Index Time / Indexing / Concepts	5,255
Index Time / Indexing / Sentences	26,480
Index Time / Indexing / Concepts Per Document	2.0258
Index Time / Indexing / Sentences Per Document	10.2082
Index Time / Indexing / Passage Concept Links	55,496
Index Time / Indexing / Concept Sentence Links	54,784
Index Time / Indexing / Sentence Passage Links	41,217
Index Time / Indexing / Embedding Dim	1,024
Index Time / Indexing / Size In Bytes	2,456,211,346
Index Time / Indexing / Size In Mb	2342.4257
Index Time / Indexing / Docs Per Second	53.995
Index Time / Indexing / Cache Loaded	no
Index Time / Indexing / Cache Path	data/processed/symfony/symphony_beir_semantic/linear_rag_cache/snow...
Index Time / Indexing / Embedding Model	snowflake/snowflake-arctic-embed-l-v2.0
Index Time / Indexing / Spacy Model	nl_core_news_sm
Index Time / Indexing / Backend	faiss
Index Time / Indexing / Phase Times / Model Load Time In Seconds	7.904
Index Time / Indexing / Phase Times / Cache Lookup Time In Seconds	0.0003
Index Time / Indexing / Phase Times / Bm25 Build Time In Seconds	0.131
Index Time / Indexing / Phase Times / Concept Extraction Time In Seconds	14.0516
Index Time / Indexing / Phase Times / Embedding Build Time In Seconds	20.0884
Index Time / Indexing / Phase Times / Cache Save Time In Seconds	4.9907
Query Time / Search / Time In Seconds	6.8462
Query Time / Search / Time Per Query In Seconds	0.1317
Query Time / Search / Queries Per Second	7.5954
Query Time / Search / Queries	52
Query Time / Search / Top K	20

Query Time / Search / Dense Candidate K	500
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Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	52
Total Pairs	1,040
Total Batches	104
Total Input Tokens	259,554
Total Wall Time Sec	4.2416
Avg Time Query Sec	0.0816
Avg Time Batch Sec	0.0408
Pairs Per Sec	245.19
Queries Per Sec	12.26
Tokens Per Sec	61193.04
Peak Gpu Memory Gb	4.38

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9199	0.8188	0.7588	0.8165	0.8565
MAP	0.2093	0.4868	0.6496	0.7139	0.7445
Recall	0.2093	0.5039	0.6843	0.8022	0.9058
Precision	0.9423	0.7885	0.65	0.3827	0.2163

Sample Top Results

Query q0: why is my webpack configuration not minifying the assets

1. Document ece57511-4f43-435e-af7e-34db3c4a4b53 | score 0.0164

[Does the AssetMapper Component Minify Assets?](#does-the-assetmapper-component-minify-assets) Nope! In most cases, this is perfectly fine. The web asset compression performed by web servers before sending them is usually sufficient. However, if you think you could benefit from minifying assets (in addition to later compressing them), you can use the [SensioLabs Minify Bundle](https://github.com/sensiolabs/minify-bundle) . This bundle integrates seamlessly with AssetMapper and minifies al...

2. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0161

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3. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 0.0159

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Sample Top Rerank Results

Query q0: why is my webpack configuration not minifying the assets

1. Document 38e1c5f2-ef8d-4104-b73e-a8b100d7029d | score 0.0904

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